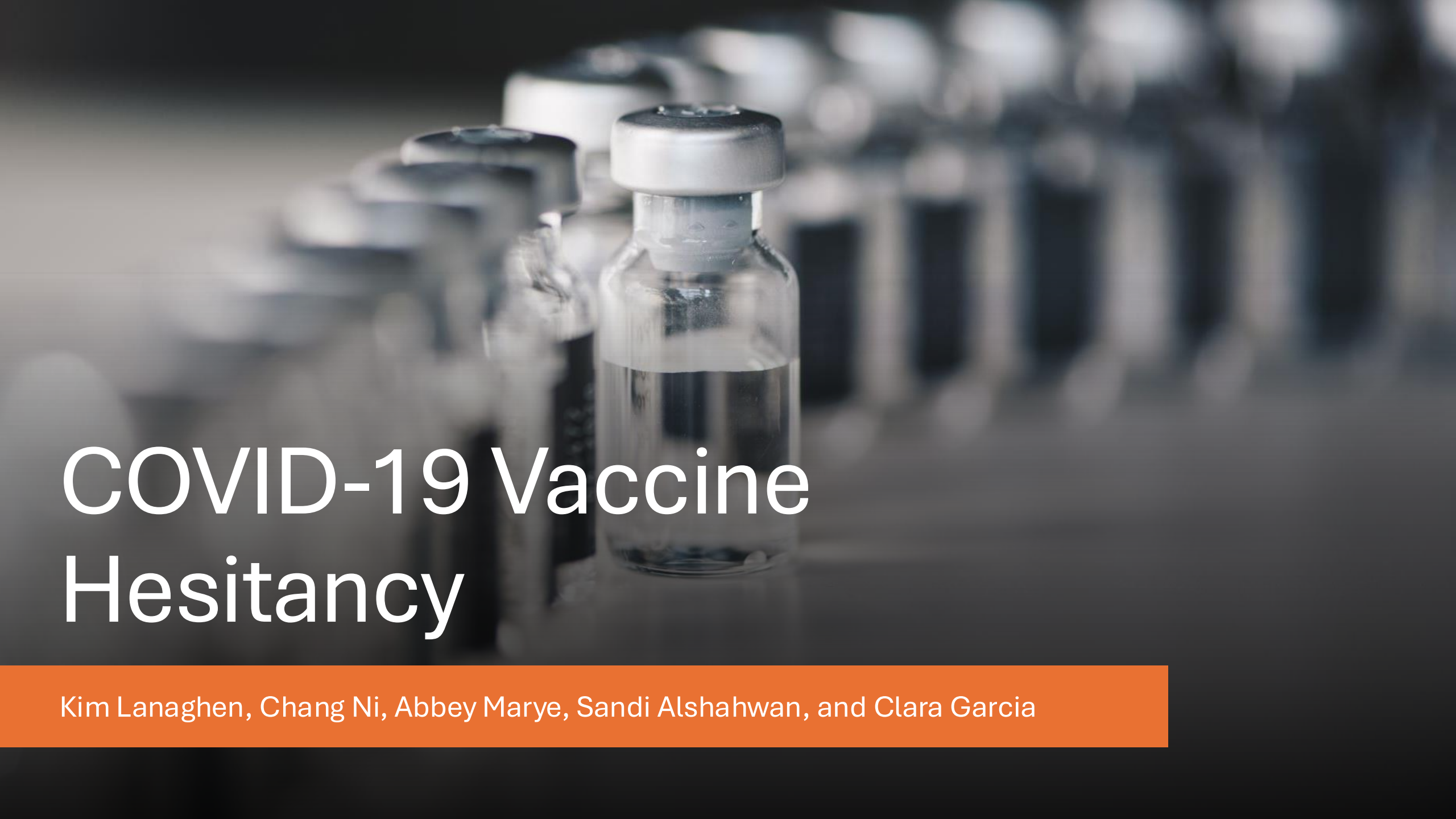

COVID-19 Vaccine Hesitancy

Kim Lanaghen, Chang Ni, Abbey Marye, Sandi Alshahwan, and Clara Garcia

Background and Motivation

- Attitudes toward COVID-19 and its vaccine have been contested since its rollout in early 2021.
- From the early stages of the COVID-19 pandemic in 2020, public responses have varied widely across geographic regions
- Media has provoked differences in risk perception, trust in public guidance, and personal health beliefs.
- Using both survey and numerical data to examine how collective beliefs and individual choices we will have a measurable effect on population outcomes.
- By linking opinion data to epidemiological trends, this project highlights how vaccine acceptance coincides with community health and well-being.
- Aim to underscore the importance of reliable information dissemination and evidence-based decision-making without making assumptions beyond what the data can support.

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Questions

- Is there data complete enough to create a new dataset to derive meaningful insights from?
- Is there a measurable correlation between vaccine attitudes and weekly COVID-19 death rates across U.S. states and counties?
- How do these relationships vary geographically and over different phases of the pandemic?
- Do Social Determinates of Health exhibit any correlation to either vaccine sentiments?
- Do regions with consistently lower vaccine acceptance exhibit different mortality trends compared to regions with high acceptance?

Objectives

To aggregate and clean data to answer further meaningful questions about COVD-19 vaccine hesitancy, SDOH, and outcomes

Answer whether individual vaccine attitudes, at a state or county level, correlate to COVID-19 death rates within a reasonable time frame (2020-2022).

To learn how individual health decisions and/or social determinants of health can affect community health on a large scale, and how this applies to disinformation, health information policy, and incentives to make well-informed decisions.

Brief analysis to observe any correlations or statistical significance between vaccine hesitancy, SDOH, and county death rates.

Data Sources

- For our project, we will use three datasets. One captures vaccine attitudes, the second COVID-19 mortality trends, and the third SDOH by county from 2017-2021.
- Vaccine Hesitancy (taken 5/27/2021-6/7/2021): https://www.kaggle.com/code/jaykumar1607/covid-vaccine-hesitancy-usa-plotly-mapbox/notebook#Demographics
- Weekly death counts by State: https://data.cdc.gov/Case-Surveillance/Weekly-United-States-COVID-19-Cases-and-Deaths-by-/pwn4-m3yp/about_data
- Social Determinants of Health by County (2017-2021):

  https://catalog.data.gov/dataset/sdoh-measures-for-county-acs-2017-2021

Data Merge

Step 1: Merge Hesitancy + Death Data

- Matched 3,140 US counties by FIPS code
- Converted raw death counts → death rates per 100,000 population
- Created both weekly and cumulative versions

Step 2: Add SDOH Measures

- Added 9 social determinants of health indicators
- Final dataset: 3,140 counties × 178 columns

SDOH Measures Included :

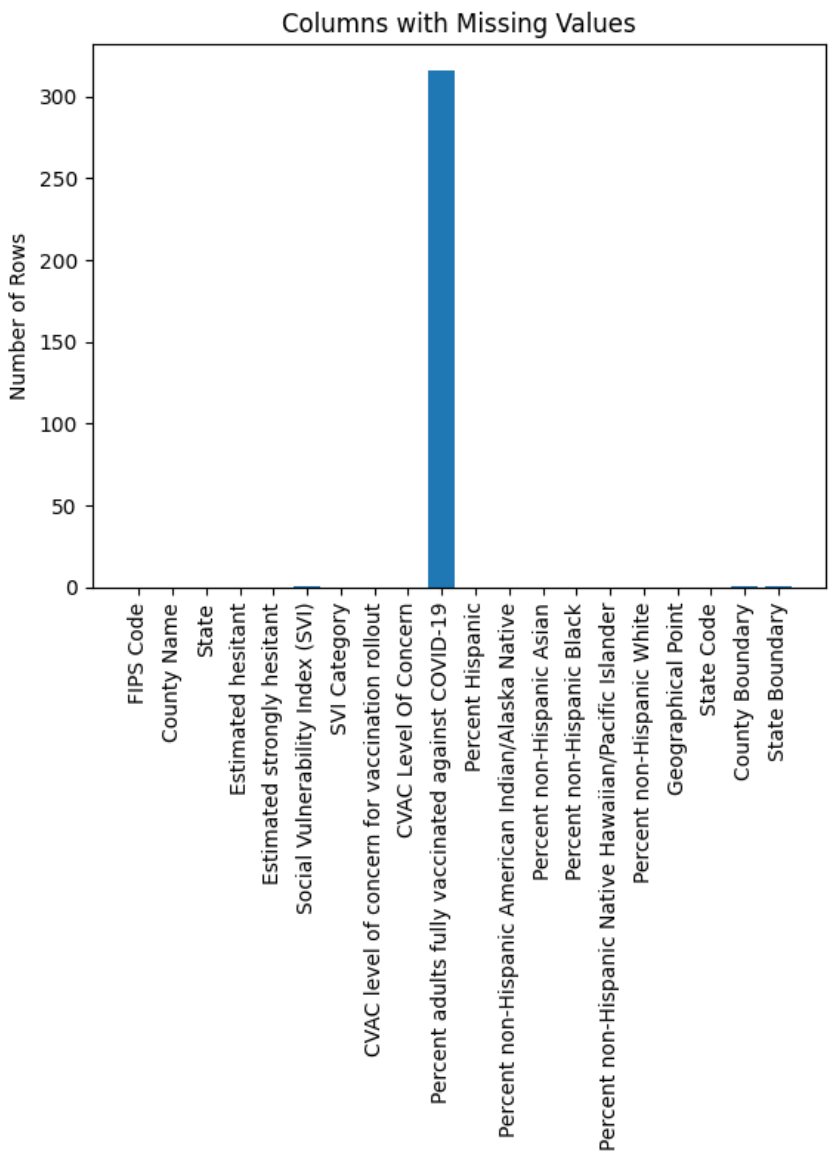
- Poverty rate, No broadband internet access, No high school diploma,
- Unemployment rate, Single-parent households, Housing cost burden,
- Crowding, Aged 65+, Racial/ethnic minority status

Data Quality Assessment

Based on Weiskopf and Weng Framework

Completeness

```
Percent missing by column:
FIPS Code                                          0.000000
County Name                                        0.000000
State                                              0.000000
Estimated hesitant                                 0.000000
Estimated strongly hesitant                        0.000000
Social Vulnerability Index (SVI)                   0.031827
SVI Category                                       0.000000
CVAC level of concern for vaccination rollout      0.000000
CVAC Level Of Concern                              0.000000
Percent adults fully vaccinated against COVID-19  10.057288
Percent Hispanic                                   0.000000
Percent non-Hispanic American Indian/Alaska Native 0.000000
Percent non-Hispanic Asian                         0.000000
Percent non-Hispanic Black                         0.000000
Percent non-Hispanic Native Hawaiian/Pacific Islander 0.000000
Percent non-Hispanic White                         0.000000
Geographical Point                                 0.000000
State Code                                         0.000000
County Boundary                                    0.031827
State Boundary                                     0.031827
dtype: float64
```

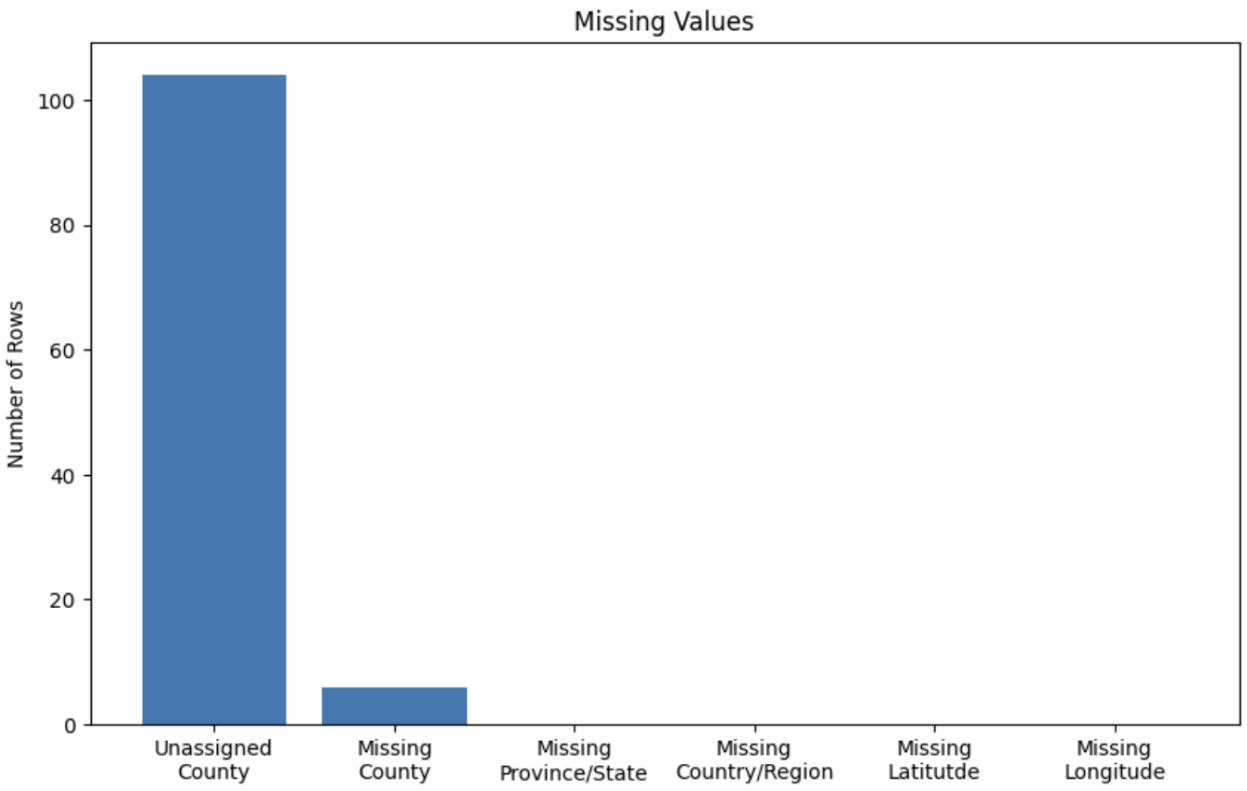


Completeness COVID-19 Deaths

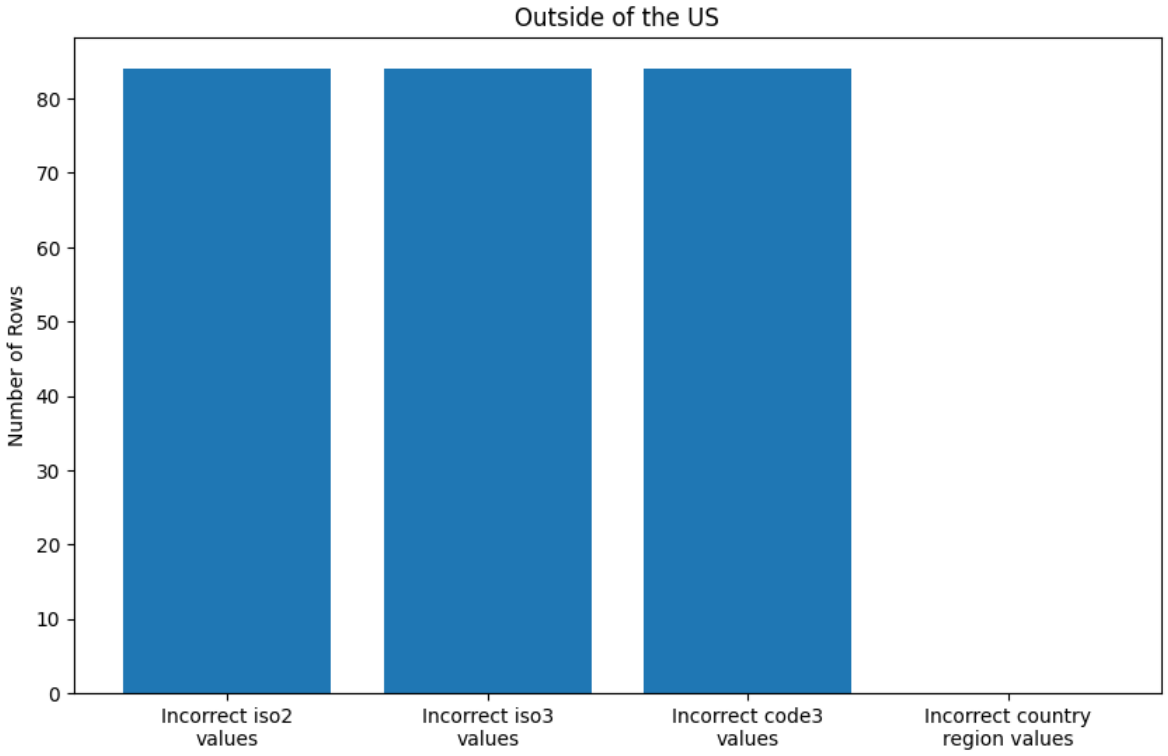
- Overall completeness: 99.9995%
- Only 10 rows missing FIPS
- 6 missing county names (Admin2)

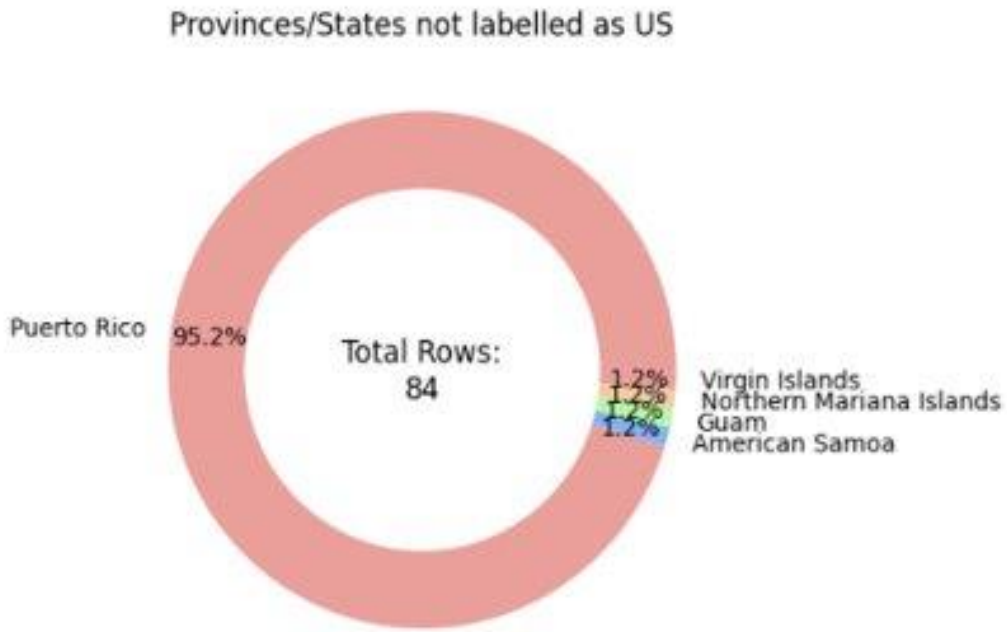
```
Missing values in Country_Region: 0

     Admin2            Province_State     FIPS
102     NaN            American Samoa     60.0
338     NaN          Diamond Princess  88888.0
572     NaN            Grand Princess  99999.0
573     NaN                      Guam     66.0
2123    NaN  Northern Mariana Islands     69.0
3009    NaN            Virgin Islands     78.0
```



Correctness COVID-19 Deaths

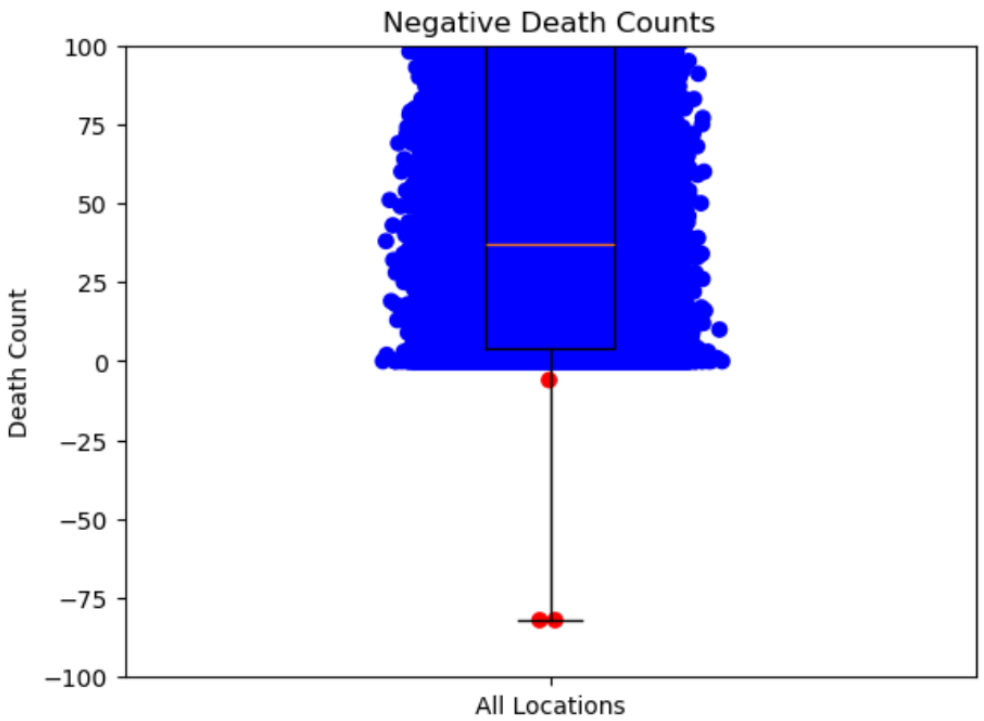




Correctness COVID-19 Deaths

- Data modification records indicate:
- "March 19, 2021: North Carolina, US | Removal of 99 deaths that did not fit the most recent version of the state definition of a COVID-19 death"(Johns Hopkins University CSSE, 2021).

- The cause of the South Carolina correction is unknown. No specific documentation was found in the JHU CSSE modification records.

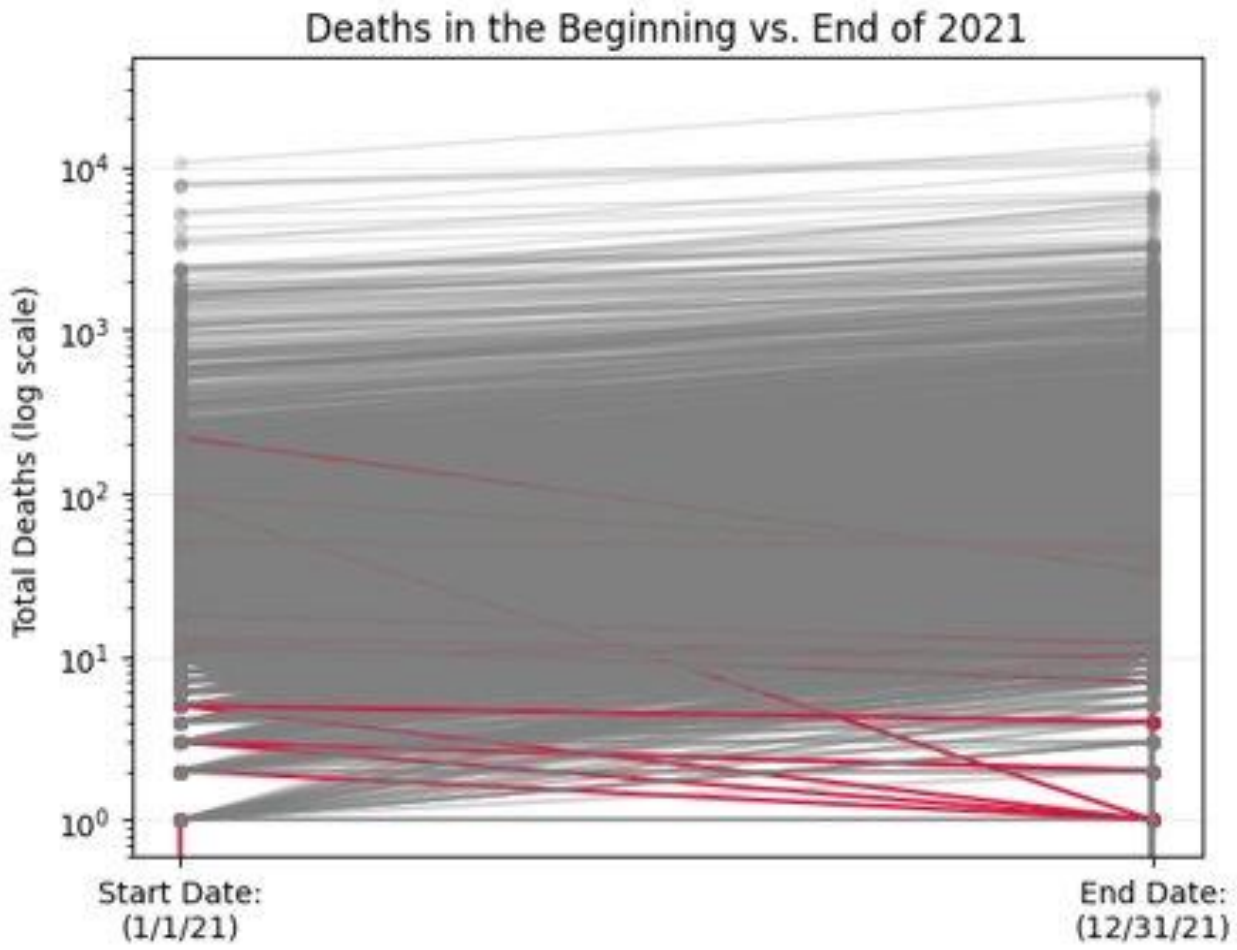


	5/5/2022	5/6/2022	11/9/2022
2056	0	0	-6
2531	-82	-82	0

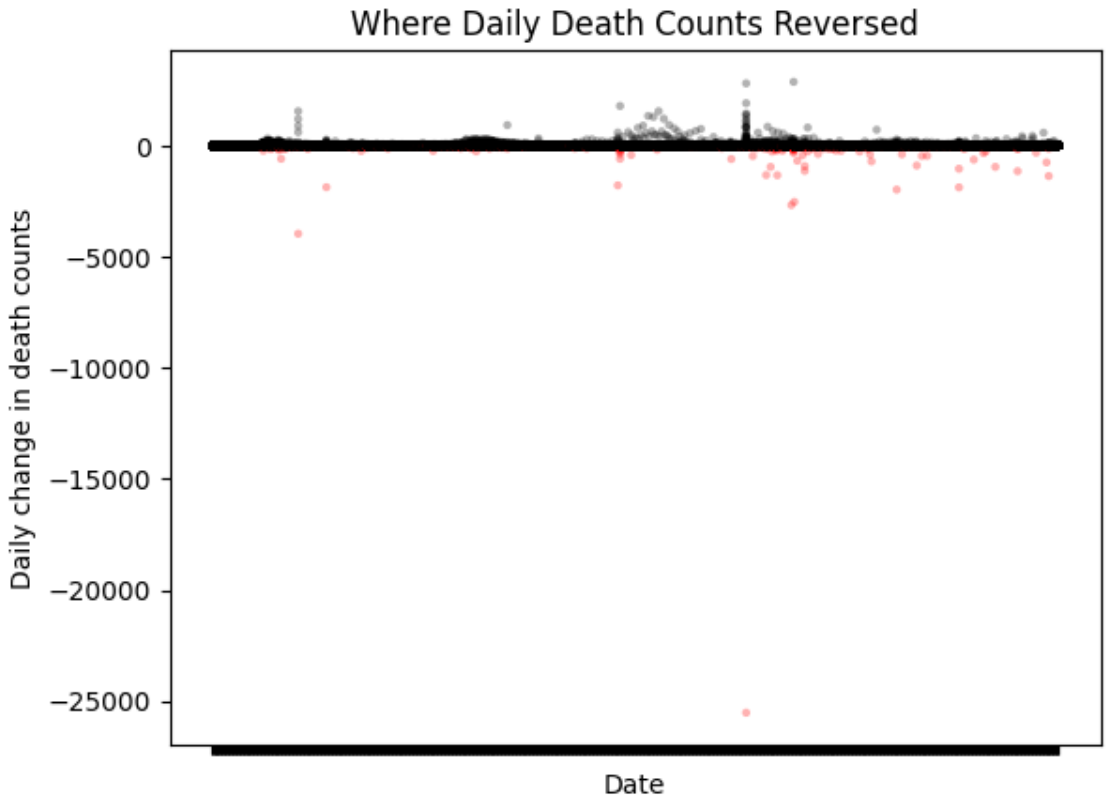
	UID	iso2	iso3	code3	FIPS	Admin2	Province_State
2056	84090037	US	USA	840	90037.0	Unassigned	North Carolina
2531	84090045	US	USA	840	90045.0	Unassigned	South Carolina

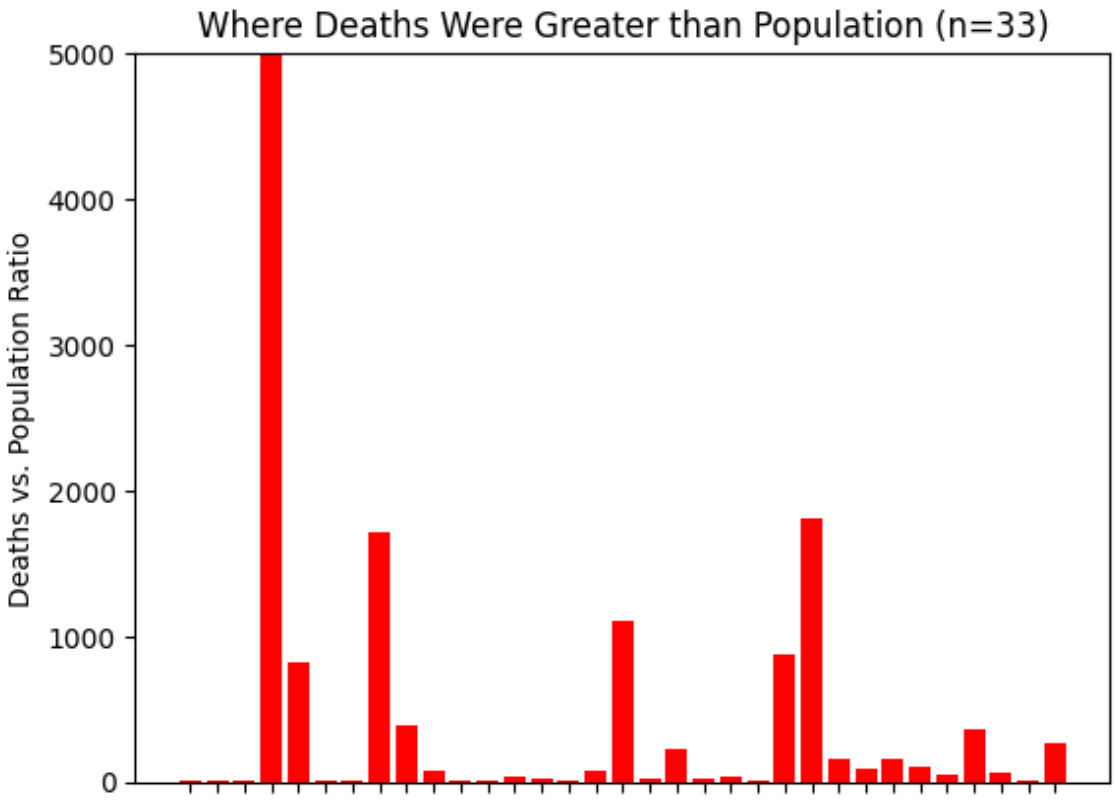
Concordance

- FIPS Codes checked to be 5 digits
- Cumulative COVID-19 death count is not expected to decrease over time
- No COVID-19 deaths before 2020
- **Counties where end of 2021 death count is less than at the start of 2021: 15**



Concordance

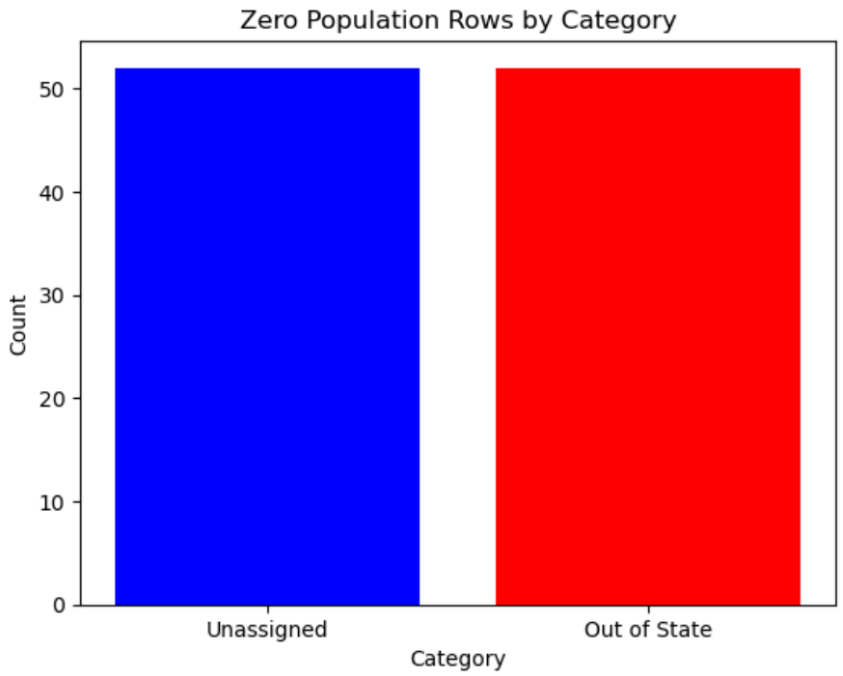




Plausibility

- No counties with less than 100,000 deaths
- 115 rows with zero population were identified
  - Correspond to 'Unassigned' and 'Out of State' placeholders rather than real locations, and are expected to have no population value

```
Total invalid population rows: 115
          Admin2 Province_State  Population
52     Out of AL        Alabama           0
64    Unassigned        Alabama           0
91     Out of AK         Alaska           0
97    Unassigned         Alaska           0
113    Out of AZ        Arizona           0
...          ...            ...         ...
3235  Unassigned  West Virginia           0
3287   Out of WI      Wisconsin           0
3306  Unassigned      Wisconsin           0
3331   Out of WY        Wyoming           0
3339  Unassigned        Wyoming           0
```



Currency

- COVID-19 deaths data was all up to date for 2021
  - Some states/territories stopped recording death counts before 2023

Results

Correlations between SDOH and Vaccine Hesitancy

Predictors:

- Poverty → +0.41
- No broadband → +0.39
- No high school diploma→ +0.28
- Single-parent → +0.19
- Unemployment → +0.16
- Housing cost burden → -0.15

Weak / not significant:

- Age 65+ → -0.06
- Minority status → +0.05
- Crowding → +0.04

Hesitancy and SDOH in Relation to Mortality

- Hesitancy significant in all models
- Betas show direction of association
- R^2 shows model strength

SDOH	Beta	R^2
No broadband	+8.47	0.26
Poverty	+7.37	0.24
No high school diploma	+9.40	0.23
Age 65+	+8.51	0.19

Conclusion

- Successfully integrated multiple datasets (hesitancy, mortality, SDOH)
- Performed data cleaning, harmonization, and quality checks
- Created a consistent, county-level analytic dataset
- Analysis showed meaningful relationships across variables
- Dataset is ready for further analysis and interpretation

References

- Centers for Disease Control and Prevention. (n.d.). *Weekly United States COVID-19 cases and deaths by state*. https://data.cdc.gov/Case-Surveillance/Weekly-United-States-COVID-19-Cases-and-Deaths-by-/pwn4-m3yp/about_data
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